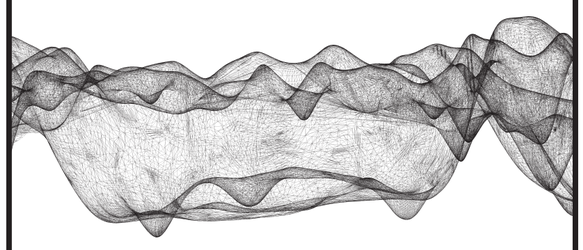
FIELDGUIDE

FIND PLAY MAKE



SPACE

FIELDGUIDE IS AN OCCASIONAL COLLECTIVE AND SOMETIMES JOURNAL FOR PEOPLE INTERESTED IN DESIGN, TECHNOLOGY AND SOCIETY AND THE INTERSECTIONS WHERE THEY OVERLAP.

OUR MOTTO IS: FIND PLAY MAKE.

IT UNDERPINS OUR WORK.
WE CELEBRATE PROCESS AND OUTCOME, UNCOVERING INSIGHTS THROUGH INVESTIGATION; PLAYING WITH IDEAS AND TECHNOLOGY TO DISCOVER NEW OPPORTUNITIES; AND MAKING ENGAGING, ENTERTAINING, AMUSING AND AMAZING ARTIFACTS THAT TELL THEIR OWN STORIES AND ENCOURAGE CONVERSATION.

THE SPACE ISSUE IS THE THIRD IN A SERIES OF OCCASIONAL FIELDGUIDE PUBLICATIONS.

START

Making data physical means that more people can access it in more ways. And as far as being able to touch data that can never be touched, then space and time have to be it.

Things that are far away or things that are lost in time are two physical barriers we simply can't cross and we want to find a way to do this.

For me, one of the most dramatic pieces of data I have come across are the chunks of wall missing from the Victoria and Albert Museum on Exhibition Road in London. I used to walk past these every day and wonder why no one had filled them in – as the V&A is one of the most precious buildings we have. Then one day I saw the small plaque where these words are carved in stone:

The damage to these walls is the result of enemy bombing during the blitz of the Second World War 1939-1945 and is left as a memorial to the enduring values of this museum in a time of conflict.

I could literally touch where shrapnel from bombs had blown holes in the building, I could touch the data. It was a far more real experience than any I had previously had of the war. The data from an event 70 years ago had touched me when I touched it. I had travelled through time.

So can we do this to space data? Can we build connections between people here on earth that reach across the vastness of space and time – to the very origins of the universe itself?

ONE OF THE MOST DRAMATIC PIECES OF DATA I HAVE COME ACROSS ARE THE CHUNKS OF WALL MISSING FROM THE VICTORIA AND ALBERT MUSEUM

We hope we can! Which is why when Ali Llewellyn from Open NASA got in touch a year ago we literally jumped for the chance to work with her and her team in making space data real here on earth. We've been hacking together examples as demonstrators, or starting points, of new ways to connect people to space data.

The question Ali posed was whether designers could turn this data into objects that encourage people to think and to dream about the wonders of space. This edition of Fieldguide features some of the results so far.

Jon Rogers

OPEN NASA

We met Ali Llewellyn, Open NASA's Community Advocate, on twitter when we were running workshops in India. Fast forward a little over a year and Jon and Ali were on a South By South West panel together running through the excitement they share in making space data physical. We asked Ali to answer a few questions for us.

FG: What excites you about making space data physical?

AL:. From the time I was a child, I wanted to touch the stars. I wanted to walk on other planets with my own feet and fly a spaceship with my own hands. While I am not an astronaut, making space data physical enables me to get closer than most humans can yet get to those opportunities. My work at NASA in open innovation and mass collaboration is dedicated to exactly this: enabling everyone on planet Earth to contribute directly and substantially to the exploration mission.

FG: What are the possibilities that this approach presents?

AL: This approach inspires everyone with the wonder of exploration by making the data engaging and allowing it to inform a new context - who doesn't want to drive Curiosity or touch the Sun? We want to extend the usefulness of space data. While the data often had one initial research purpose, we are "recycling" it for other applications and uses, especially in new contexts.

FG: Why is it important for us to make things from data?

In the time it took you to read this sentence, NASA gathered approximately 1.73 gigabytes of data from our nearly 100 currently active missions! We do this every hour, every day, every year – and the collection rate is growing exponentially.

IN THE TIME IT TOOK YOU TO READ THIS SENTENCE, NASA GATHERED APPROXIMATELY 1.73 GIGABYTES OF DATA FROM OUR NEARLY 100 CURRENTLY ACTIVE MISSIONS!

Data is one of our most valuable assets. We are committed to making our data as accessible as possible, both for the benefit of our work and for the betterment of humankind through the innovation and creativity of the over seven billion other people on this planet who don't work at NASA. What would become possible if everyone could not just access but remix and reuse the images, maps, metrics and lessons learned from this amazing trove of observation?

MINI MARS ROVER



The Mars Curiosity rover sealed its fate when it took a one-way ticket to Mars, where it bravely explores Martian landscapes all alone, while we live in comfortable companionship down on earth.

Can we somehow connect this lonely, intrepid robot to our homes, letting us cheer it on and know our thoughts are with it as it roams the unknown?

It's easy to forget about something so far away. For our empathy to be triggered, we need reminded what the Mars Curiosity is up to. We could look at the data it produces, but for many of us, statistics are not emotive.

What would our response be if we had something physical in our home that made this data feel more real?

Roll in the Mini Mars Rover… built for NASA's International SpaceApps Challenge, it is an Electric Imp Internet connected robot that mimics the movements of its big brother.

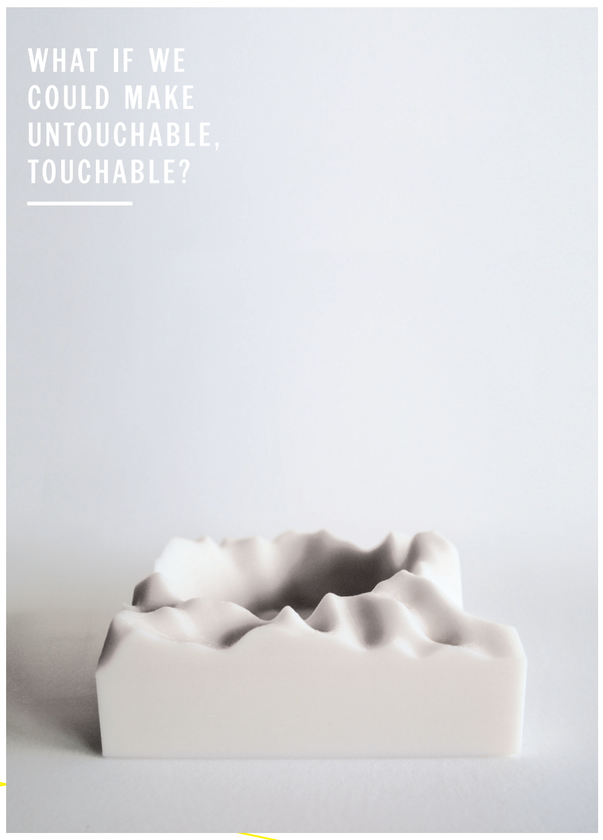
As the Mars Curiosity rover moves, the Mini Mars Rover follows the same pattern, and streams live images and audio from Mars. By manifesting boring and hard to find data in this little robot, we are able to make a heap of statistics tangible, accessible and engaging.

The Mini Mars Rover was designed to excite space enthusiasts, but more importantly to ignite curiosity about space in anyone and everyone. It is far easier for people to have an emotional connection with a physical object than data, especially if the object is driving about their home, bashing into walls and getting into mischief.

Our hope is that as they run to stop the Mini Mars Rover from crashing, there will be a moment of thought given to the Mars Curiosity rover, potentially the only life on Mars, battling with its mission of discovery.

Mike Shorter

TOUCHING THE MOON



Space is big. And far away. It is, except to a very small group of people, untouchable.

Physicists and engineers of many stripes have, over the years, made sure that it is not unknowable, but for most it remains out of reach. It is this distance that is behind much of the mystique of space; it is why it inspires. It is a new frontier for us to explore; something so vast that we can imagine almost anything is possible out there.

It is this sense of scale that I feel is where the awe that we have for space comes from – the distances are so huge, the objects so large, that we look out upon the stars at night in a similar way to a little child seeing a jumbo jet up close for the first time. We simply can't take in the scale, and are left with little to do but stare in wonder.

But what if we could make untouchable touchable? What could we do with 'bits of space'?

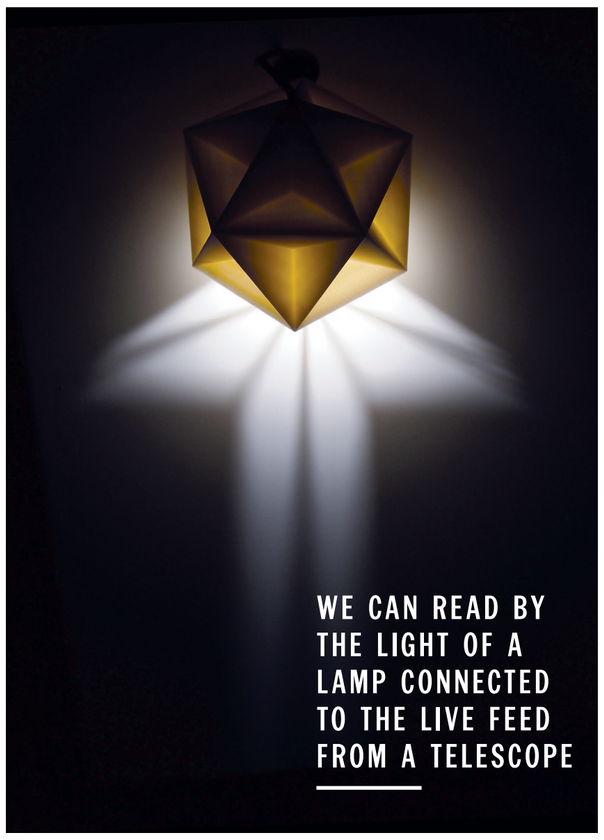
I posed this question to some of my students recently and we hit upon an idea that would allow school pupils to carry out experiments on bits of the moon in their classrooms. Our plan was to 'print the moon' making use of modern 3D printing technology and data from the NASA Lunar Reconnaissance Orbiter Satellite.

You can use shadows cast by the model to make estimates of the dimensions of craters, all from the comfort of a lab bench. More than that, because the satellite data covers the whole of the Moon's surface we were able to print out a copy of a bit of the dark side of the moon for added coolness.

As the new Scottish Higher and Advanced Higher will include much more material on space and astrophysics, I hope that by making space data physical we can play a little role in helping teachers to inspire the next generation of scientists, engineers and explorers.

David McGloin

TIME & SPACE



As part of our SXSW and Edinburgh Science Fair presentations, we wanted to make things that gave data a physical presence.

We asked Jayne Wallace, a contemporary digital jeweller, to get involved. Jayne's group made experience prototypes of new ways to connect us to space and time: A light that flickers in response to live star data, a wind chime that blows in a solar wind and a rough cut diamond that casts a shadow of its true potential when the earth is aligned with the Diamond Planet Cancri 55 e. This is Jayne on Time and Space.

Our lives have a pace to them and time is both something we crave more of, yet know has an ultimately finite quality for us. Our interactions with the digital are quickening our pace of life, altering not only the texture of days and years but also how we value, measure and perceive the passing of time.

But there are things that are bigger than us, older than we can imagine, that give our atomised view of life and the time we have a very different perspective; we simply have to look up to engage them.

We want to explore what it would mean to use digital technologies and space data to subvert our relationship with time and bring fresh potential to the digital objects we live with and through.

We can create ways to experience things that occurred before humankind existed; we can read by the light of a lamp connected to the live feed from a telescope and know that when it flickers, a new planet has been discovered.

We can connect to the orbits and rhythms of planets through objects that gently respond to these different cycles and be reminded that we are part of something much greater, much faster, much slower and much more fascinating than our atomised lives sometimes allow us to consider.

Jayne Wallace

MOON GOLF



Like many men of my age, I wanted to be an astronaut when I grew up. 40 years later, I'm still waiting to grow up and I still want to go to the moon.

What better use of my remaining time and faculties than to become the first human being to play a full round of golf on the moon? And I'm not talking here about the three rushed practice swings that Alan Shepard managed. This would be a full 18-hole course built, like so many golf courses these days, on a beautiful pristine unspoiled site of special scientific interest.

I've already been meeting up with various people over the past few years to walk, talk and pitch a few balls around designated areas here on earth. In each case, I have studied the terrain very carefully and made recommendations for a teeing-off point, a 'green' and a 'hole'.

My fellow players and I then walk and play across the designated course (often spaces where playfulness is generally frowned upon - city streets, libraries, golf courses, etc.) sharing with each other our dreams, ambitions and thoughts about how we all might play our way into the future.

When we've finished each outing, I use all the data about where we've been, each shot taken, where we have finally holed out etc., and find an equivalent place on the surface of the moon where this hole could be laid out and played. It's actually quite spooky to note how readily our earthly practice rounds can be mapped onto the moon. It gives me quite a lot of hope that one day my dream can come true, that everything we're doing here now on earth can eventually be played out in space in the future.

Already we have half a dozen holes that have been walked and played both here in the UK and abroad. If anyone out there is inspired by this idea and would like to contribute to the making and mapping of more holes, please do get in touch.

Tim Wright

JOURNALISM & DATA

Those of you who have picked up and read The Insight Journalism issue of Fieldguide will have met our friend, collaborator and fieldguide partner, the journalist and media innovator Paul Egglestone.

Representing the far-reaching impact of space and space data, we wanted to hear what a journalist would make of what we're doing by making space data physical.

We asked him if there was any value beyond the tangible beauty that physical data can provide. This is what Paul has to say.

Olivia Newton John first invited us to 'get physical' back in the 1980's and now, as illustrated through the work shown in this edition of Fieldguide, making data physical is becoming a contemporary obsession.

In journalism, 'hipster' hacks are looking for ways to create stories from the ones and zeros languishing on computer databases.

Some of this code is available to anyone with a computer, Internet connection and the technical skills and time to collate the data, analyse it and then create stories based on their interpretation. Data Journalism, as it is known, is now synonymous with the production of infographics. Hans Rosling and David McCandles have educated us about poverty and public spending through elaborate graphical representations that attempt to 'show' rather than 'tell' us about issues.

Online, these presentations are interactive, enabling viewers to create new stories as they manipulate them. This way of making data physical in order to give alternative perspectives is now offering a third dimension to data journalists and their audience. Literally.

Moving off screens and into the real world creates a whole new paradigm of storytelling through physical objects responding to data – and you've got to ask what space data journalism should really look like.

Paul Egglestone

IMAGES

The reverse shows a mosaic view of the Moon's north pole assembled from 18 images taken by Galileo's imaging system through a green filter as the spacecraft flew by on December 7, 1992. The Moon's north pole is located just inside the shadow zone, about a third of the way from the top left of the illuminated region. © NASA

The cover shows a wireframe model of the data used by David's team to print a piece of the moon.

FIELDNOTES

Kepler
was a 17th Century mathematician and astronomer who laid down the foundations of much of what we know as astronomical physics today.

The Kepler observatory
is a spacecraft that NASA built to discover earth-like planets orbiting stars.

Hacking
is a way of exploring problems through playing rather than officially sanctioned problem solving. In the summer of hack in 2012, there were thousands of global hack events – from NASA to Mozilla, from government to cultural oganisations covering subjects as diverse as the future of news to the understanding of healthcare.

A Maker
is a person who makes things. As we head beyond the loaded language of nouns to describe what we do and move towards verbs we are able to give ourselves names that reflect what we want to call ourselves.

Open Data
is used to describe the philosophy and practice of making datasets freely available for people to use, manipulate and re-publish without restriction or high-level control.

Visualisation
is an approach to presenting information clearly and in a visual format. Traditionally this has meant representing information through graphical means in our definition we apply this in a wider visual sense to include objects, models and arfefacts.

Electric Imp
A small SD card sized programmable device that can connect to the internet allowing physical objects to pull and push data to and from the internet.

Sivilization
the life Huck Finn tries so desperately to escape. It is tradition, conformity, education, conservatism, capitalism, oppression, constraint, security, warmth, comfort and old-fashioned community.

WHAT IF?

We asked our friends at NASA to take a moment to reflect on what they do and what inspires them about space. They tell a story of ambition and dreams that go to the very core of what makes us look up at night and ask what if…

I never really understood the concept of infinity. Everything on earth is finite. Los Angeles to San Francisco is 2,488 miles. There are 24 hours in a day. The speed limit is 65 miles per hour. In space, this all changes. Everything becomes relative. Scale increases at an unimaginable pace.

How can we, a species so used to being masters of our domain, of having control over almost every aspect of our planet, possibly comprehend the fact that, in reality, we are an insignificant planet among trillions and trillions of equally insignificant planets, in a universe that goes on and on and on for eternity?

Every time I look up at the night sky that question races through my mind. I don't believe our thirst for understanding the universe will ever be fully satisfied, but the speed at which we are learning is nothing short of astonishing.

Within our lifespans, it is likely that humanity will become a multi-planet species. Although it seems like technology moves at a glacial pace, in reality we are accelerating faster than ever before. The future is an exciting place.

Sean Herron
Open NASA

Space is the one cool place that hasn't been explored by everyone and their mom. There is so much out there that we barely know anything about. The idea is fascinating to me, and why I decided to intern with NASA Johnson Space Center. Being here at NASA, my interest has grown even more. I am constantly learning about new inventions and ideas pertaining to space flight. I get to job shadow brilliant engineers and scientists, as well as learn from my colleagues on a daily basis. Educating the public about what we do here is huge in the Education Department.

We want people to understand how cool the stuff we're working on is, and that it's not a waste of time. Coming on site everyday and exploring all the cool buildings, using my "official" badge, and congregating with brilliant people motivates me to contribute all I can to NASA.

Madison Wright
Johnson Space Center

Every person on this planet has something of value to our endeavor to explore. When we land on Mars, I'll look at the stars, reflect on our achievement, and wonder what's next.

Kaylin Turner
Johnson Space Centre

What inspired me about space was the constant variable of the unknown.

When you just think of the limitless amount of questions that can be asked about space... it's amazing what could be out there!

Also, the benefits of exploration that we are already witnessing, from the biological to technical, are helping people everywhere.

Haley Stephenson
NASA Headquarters

What personally drew me to work for the space industry is the diversity of skills and personalities it takes to make a mission successful. Our first thoughts when we ponder the space program are often about the astronauts; what follows is usually some sort of acknowledgement of the impressive engineering feat that building a spacecraft is. Rarely do we think of the men and women who train these astronauts, the people who found the money to support the project, the businessmen, the writers, the publicists, much less the doctors, biologists… ordinary people contributing to absolutely extraordinary things.

Space is not just a vacuum of death, radiation and rocks hurtling towards us… it is a challenge! We in the space industry are the pioneers, the dreamers and explorers of the universe. We subscribe to ambitions greater than ones we can find here on Earth.

Generations might pass and we will see many new faces before we put a man on Mars, but so long as the space industry has those who dream we will see the expanses of this galaxy.

Maybe I have fallen for the romantic appeal of the job and am just too satisfied being among some of the greatest minds that the world has ever known.

Samuel Redus
Johnson Space Center

SIVILIZING SPACE

What if we paused in our space quest and looked back on earth for a moment? To help us, we asked the writer and contemporary English scholar, Jim Byatt, to give us a view from literature.

Terrestrial space, in case you hadn't noticed, is in short supply, and becoming more so with each generation. Science fiction certainly noticed. The plight of the urban dweller, an evolutionary by-product of the industrial age, reaches a most unpalatable climax in Harry Harrison's Make Room, Make Room, and is scarcely more appealing in JG Ballard's "Billennium" (though the food's better). Anyone who thinks these stories are pure fantasy needs only to look back over the past century or so at the rise of vertical habitats to realise that we have progressed, almost unconsciously, to a state where we are living quite literally on top of one another. Some of the world's taller skyscrapers can house tens, even hundreds of people on a geographical footprint no larger than the average vegetable allotment.

To counteract the inevitable consequences of intolerable overcrowding, a condition loitering ominously in the not-so-far future, we're rapidly becoming conversant with other forms of space. Cyberspace, the virtual world whose boundaries are indefinable, has served to expand our horizons in order that, even in the confines of a restricted living environment, we can fool ourselves into thinking we are running free. Cyberspace has brought about two paradoxical phenomena. On the one hand, the world has been shrunk by a previously unimaginable factor. If the birth of the railway made us rethink the notions of time and space, the birth of the computer has all but eradicated them. Through chatrooms, online gaming and Facebook, Marshall McLuhan's Global Village has, essentially, been realised. With Skype, we can not only communicate with people on the other side of the world in real time, we can also look them in the eye as we do it. The world has become an awful lot smaller.

Yet at the same time, this erasure of spatial distance has been accompanied by a curious corollary, the introduction of a virtual diaspora which threatens local communities and nuclear families in unprecedented ways. With each successive generation, we are becoming more naturalised to, and more dependent upon, virtual channels of communication, virtual daily habitats (including online shops, social spaces and workplaces), and virtual identities, none of which involve any physical displacement.

What does this do for local relationships? Are friends really friends anymore? What does "society", big or small, mean now? As we light out, as Huck Finn did, for the (cyber)territory, what are we leaving behind? Sivilization?

Jim Byatt

THANKS

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Jon Rogers
Mike Shorter
Pete Thomas

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NEXT

What's next? We don't know exactly, but we do know what we're interested in and the themes we'd love to explore. We were inspired by our trip to India and working with the Unbox Festival Team; we've been getting more and more involved with hacking and high-speed making; and as ever we have a passion for paper, from traditional paper making to high-tech possibilities of paper electronics. Perhaps we'll see you at the next event?

Get in touch at www.findplaymake.com.

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